

# GATE All Branch (Hinglish)



General Aptitude

**Quantitative Aptitude**

**DPP 11 Discussion Notes**  
Time & Work



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# MCQ

If 72 men can build a wall 280m. long in 21 days, how many men will take 18 days to build a similar type of wall of length 100m.?

- ☒ A 30
- ☐ B 10
- ☐ C 18
- ☐ D 28

$$\begin{aligned}
 & \overset{2}{\cancel{36}} \times \overset{5}{\cancel{100}} \times \overset{3}{\cancel{21}} \\
 & \cancel{72} \times \frac{\cancel{100}}{\cancel{280}} \times \frac{\cancel{21}}{\cancel{18}} \\
 & \quad \quad \quad \begin{matrix} 4\phi \\ 2 \\ 1 \end{matrix} \\
 & = \underline{\underline{30}}
 \end{aligned}$$

| Men | Wall | Days |
|-----|------|------|
| 72  | 280m | 21   |
| ?   | 100m | 18   |



# MCQ



A takes twice as much time as B or thrice as much time as C to finish a piece of work. Working together, they can finish the work in 2 days. B can do the work alone in

- A** 12 days
- B** 4 days
- C** 8 days
- D** 6 days

$$A = 6x; B = 3x; C = 2x$$

$$A = 2B = 3C \text{ (Efficiency)}$$

$$\frac{1}{6x} + \frac{1}{3x} + \frac{1}{2x} = \frac{1}{2}$$

$$A : B : C = 1 : 2 : 3$$

$$\frac{6}{6x} = \frac{1}{2}$$

$$\text{Days } A : B : C = 6 : 3 : 2$$

$$x = 2$$

$$3 \times 2 = 6$$



# MCQ

A contractor undertook to finish a certain work in 124 days and employed 120 men on it. After 64 days, he found that he had already done  $\frac{2}{3}$ <sup>rd</sup> of the work. How many men he can discharge now so that the work may finish in time.

- ☐ A 24
- ☒ B 56
- ☐ C 64
- ☐ D 80

$$\begin{array}{l}
 \cancel{120} \times \frac{64}{\cancel{60}} \times \frac{1}{\cancel{3}} \times \frac{\cancel{3}}{2} = \underline{64} \\
 \text{Men} \quad \quad \quad \underline{120} \\
 \quad \quad \quad \quad ? \\
 \text{Days} \quad \quad \quad 64 \\
 \quad \quad \quad \quad 60 \\
 \text{Work} \quad \quad \quad \frac{2}{3} \\
 \quad \quad \quad \quad \frac{1}{3}
 \end{array}$$

$120 - 64 = \underline{56}$

# MCQ

A can do  $\frac{3}{4}$ <sup>th</sup> of a work in 12 days. In how many days can he finish  $\frac{1}{8}$ <sup>th</sup> of work?

- ☐ A 1 day
- ☒ B 2 days
- ☐ C 4 days
- ☐ D 8 days

$$\frac{4}{12} \times \frac{1}{8} \times \frac{2}{\frac{4}{3}} = \underline{\underline{2}}$$

| Work          | Days |
|---------------|------|
| $\frac{3}{4}$ | 12   |
| $\frac{1}{8}$ | ?    |



# MCQ

Peter does 75% of work in 12 days. He then calls Charlie for help and they both complete the rest of the work in 3 days. How many days would Charlie have taken to complete the work alone?

- ☐ A 18 days
- ☐ B 24 days
- ☐ C 72 days
- ☒ D 48 days

Peter

75% → 12

100% →  $12 \times \frac{4}{3} = 16$

Charlie → 'x' days

$P = \frac{1}{16}$      $C = \frac{1}{x}$

$\frac{1}{16} + \frac{1}{x} = \frac{1}{12}$

$\frac{1}{x} = \frac{1}{12} - \frac{1}{16} = \frac{1}{48}$

48

P + Charlie → 25% → 3

100% → 12



# MCQ



If A is twice as good workman as B and therefore is able to finish a job in 40 days less than B, how many days will it take to finish the same job if A and B work together?

$$2x - x = 40$$

$$A = x \text{ days}$$

$$B = 2x \text{ days} \quad x = 40$$

$$\boxed{\frac{1}{x}}$$

$$\boxed{\frac{1}{2x}}$$

$$\frac{1}{40} + \frac{1}{80} = \frac{3}{80} \Rightarrow \frac{80}{3} = 26\frac{2}{3} \text{ days}$$

**A**  $28\frac{1}{2}$  days

**B** 40 days

**C**  $26\frac{2}{3}$  days

**D** 22 days



# MCQ



Worker A alone can do a piece of work in 6 days and B alone in 8 days. A and B undertook to do it for ₹4000. With the help of worker C, they completed the work in 3 days. How much money will be given to C?

$$A = \frac{1}{6} ; B = \frac{1}{8} ; C = \frac{1}{x}$$

$$A:B:C = 6:8:24$$
$$= 3:4:12$$

$$\frac{1}{6} + \frac{1}{8} + \frac{1}{x} = \frac{1}{3}$$

$$C = \frac{1}{8} \times 4000$$
$$= 500$$

Efficiency

$$\Rightarrow \frac{1}{x} = \frac{1}{3} - \frac{7}{24} = \frac{1}{24}$$

A:B:C

$$= 48:36:12$$

$$= 4:3:1$$

**A** ₹ 500

**B** ₹ 350

**C** ₹ 400

**D** ₹ 600



# MCQ



A and B can do a job together in 7 days. A is  $1\frac{3}{4}$  times as efficient as B. How long does it take for A to do it alone?

A  $9\frac{1}{3}$  days

**B 11 days**

C  $15\frac{1}{2}$  days

D  $17\frac{1}{3}$  days

$$A = 4x, B = 7x$$

$$\frac{1}{4x} + \frac{1}{7x} = \frac{1}{7}$$

$$\frac{11}{28x} = \frac{1}{7}$$

$$x = \frac{11}{4}$$

$$A : B = \frac{7}{4} : 1 \quad (\text{Efficiency})$$

$$A : B = 1 : \frac{7}{4} \quad (\text{Days})$$

$$= 4 : 7$$

$$A = 4 \times \frac{11}{4} = \underline{\underline{11}}$$

# MCQ

A and B can do a work in 10 and 12 days. They start the work and B leaves after three days. If daily wages are Rs. 20 for each how much does A get?

**A** 150

**B** 90

**C** 100

**D** 130

$$7.5 \times 20 = 150$$

$$A = \frac{1}{10} \quad B = \frac{1}{12}$$

$$\frac{x}{10} + \frac{3}{12} = 1$$

$$6x + 15 = 60$$

$$6x = 45$$

$$x = \frac{45}{6} = 7.5$$



**MCQ**

12 men can do a work in 15 days working 8 hours a day. In how many days can 9 men do the same work, working 10 hours a day?

- |   |    |
|---|----|
| A | 10 |
| B | 16 |
| C | 18 |
| D | 24 |

$$\frac{15}{15} \times \frac{4}{12} \times \frac{8}{10}$$

$z = 16$

| Men | Days | hrs/day |
|-----|------|---------|
| 12  | 15   | 8       |
| 9   | ?    | 10      |



# Thank You!

GW Soldiers